

COM S 2300 HW6

1a) $14 - 5 = 9$
 $23 - 14 = 9$
 $32 - 23 = 9$

Hence, we can say that the difference between 2 consecutive terms is 9.

Let a, d and a_n be the first term, the difference between 2 terms and the term at position n :

$$a = 5$$
$$d = 9$$

Thus we can say that the base case is 5 and the recursive rule is $a_n = a_{n-1} + 9$.

b) We can breakdown the terms in the following way:

$$13 = 4 \cdot 3 + 1$$
$$40 = 13 \cdot 3 + 1$$
$$121 = 40 \cdot 3 + 1$$

Thus, we can say that in this sequence each term is the sum 3 times the previous term and 1.

Hence, we can say that the base case is 4, since it's the first term and the recursive rule is $a_n = 3a_{n-1} + 1$.

2) **Prove:** $f_0 + f_2 + f_4 + \dots + f_{2n} = f_{2n+1} - 1$

We can define the fibonacci sequence as follows:

$$f_0 = 0, f_1 = 1, f_n = f_{n-1} + f_{n-2}$$

Base Case

For $n = 0$:

$$f_0 = 0$$

R.H.S:

$$f_{2(0)+1} - 1 = f_1 - 1 = 1 - 1 = 0$$

Thus, we can say that L.H.S = R.H.S and the equation holds true for $n=0$.

Assume: Let us assume that this statement holds true for some arbitrary k, that is,

$$f_0 + f_2 + f_4 + \dots + f_{2k} = f_{2k+1} - 1$$

Prove: $f_0 + f_2 + f_4 + \dots + f_{2k} + f_{2(k+1)} = f_{2(k+1)+1} - 1$

L.H.S = $(f_0 + f_2 + f_4 + \dots + f_{2k}) + f_{2k+2}$

According to our assumption:

$$(f_{2k+1} - 1) + f_{2k+2} \quad (1)$$

From the fibonacci sequence, we can say that:

$$f_{2k+3} = f_{2k+2} + f_{2k+1}$$

Thus we can write (1) as:

$$(f_{2k+1} + f_{2k+2}) - 1 = f_{2k+3} - 1$$

Thus, by mathematical induction, we can say that $f_0 + f_2 + \dots + f_{2k} = f_{2k+1} - 1$.

3) Prove: After n steps we will end up on step 4 or step 0, if and only if n is divisible by 4.

Transitions: Let us state the smallest transitions from state 0 and state 4.

State 0:

- $0 \rightarrow 1$
- $1 \rightarrow 2$
- $2 \rightarrow 3$
- $3 \rightarrow 0$

This cycle forms: $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 0$

State 4:

- $3 \rightarrow 4$
- $4 \rightarrow 5$
- $5 \rightarrow 6$
- $6 \rightarrow 7$
- $7 \rightarrow 0$ or $7 \rightarrow 4$

This cycle forms: $3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 7 \rightarrow (0, 4)$

Thus, by looking at the transitions above we can say that we need to strengthen our induction hypothesis because a direct induction proof does not include the cycles that involve 4 and 7.

Hence, we can redefine our hypothesis as follows:

For all $n \geq 0$, we can say that, if we take n steps from 0:

- We will be in a state 0 if n is a multiple of 4
- We will be in a state 4 if $n \bmod 4$ is 2

Base Case:

$n = 0$

$n = 1$, we move from $0 \rightarrow 1$

$n = 2$, we now move on from $0 \rightarrow 1 \rightarrow 2$

$n = 3$, we now move on from $0 \rightarrow 1 \rightarrow 2 \rightarrow 3$

$n = 4$, we now move on from $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4$

Therefore, we can say that the base case holds true.

Assume: Let us assume that if k is divisible by 4, then we end at state 0 or if $k \bmod 4$ is 2, then we end at state 4.

Prove: $P(k+4)$ holds true

Since we know that a cycle's length is 4, if we take 4 more steps from a valid case, we must return back to states 0 and 4. So that also means that if at k , we are at state 0, then if we take 4 steps, then we will end up at state 0 again. Similarly, if at k , we are state 4, then if we take 4 more steps, then we will end up at state 4.

Hence, we have completed the induction by strengthening our thesis.

Therefore, by mathematical induction we have proven that starting from state 0, we will end up on state 0 or state 4 after n steps if n is divisible by 4.

4) Prove: $P(n)$ is true for all positive integers n , where n can be written as a sum of distinct powers of 2.

Base case: $n=1$

For $n = 1$, we have:

$$1 = 2^0, \text{ which is a sum of distinct powers of 2.}$$

Assumption: Let us assume that the statement holds true for all integers up to k , where, $1 \leq m \leq k$, can be written as a sum of distinct powers of 2, for every integer m .

Prove: We need to prove that $k+1$ can also be expressed as a sum of distinct powers of 2.

Let 2^j be the largest power 2 such that $2^j \leq k + 1$ and let m be the difference between $k + 1$ and 2^j , such that $0 \leq m < 2^j$.

Thus, by inductive hypothesis, $m \leq k$, so according to our assumption, we can express m as a sum of distinct powers of 2.

Hence, if we add 2^j to m , it gives us a sum of distinct powers of 2 for $k+1$.

Therefore, by strong induction we have proven that every positive integer can be written as a sum of distinct powers of 2.

5) The robot basically starts at (1,2) and it can take the following steps:

- (x+1,y-4)
- (x-1,y-2)
- (x+4,y-1)
- (x+3,y)

Over the states are the (x,y) co-ordinates and transitions are governed by 4 step rules

Prove: Robot cannot reach (0,1)

Let us first try and analyze the changes in x and y .

Move	Δx	Δy
(+1,-4)	+1	-4
(-1,-2)	-1	-2
(+4,-1)	+4	-1
(+3,0)	+3	0

If we take a look at the transitions of x , then we can say that +1,-1,+4,+3, so that means it moves from odd to even or comes back to odd, whereas, y changes by -4,-2,-1 and 0 so that means it remains even or becomes odd.

Let us determine whether the sum of each move is odd or even.

Initial step: $(1,2) = 1+2 = 3$ (odd)

$(+1,-4) = +1-4 = -3$ (odd)

$(-1,-2) = -1-2 = -3$ (odd)

$(+4,-1) = 4-1 = 3$ (odd)

$(+3,0) = 3+0 = 3$ (odd)

Thus, we can say that the sum of all transitions is an odd number.

Now, the target state is $(0,1)$, where the sum of the digits is also odd (1), however according to the question we have to check whether it is always odd for any reachable state or not.

In order to prove this, let us check whether $(0,1)$ is produced via valid transitions or not.

Let's make a check for previous possible states for $(0,1)$:

- $(-1,3)$, which comes from $(-1,4)$, is not possible since $x = -1$ is not reachable
- $(1,3)$, which comes from $(-1,-2)$ is not possible since $(1,3)$ never occurs.
- $(-4,2)$, which comes from $(+4,-1)$, is not possible since $x = -4$ is not reachable
- $(+3,1)$, which comes from $(+3,0)$, is not possible since $x = -3$ is not reachable.

Hence, we can say that $(0,1)$ cannot be reached because the sum $x+y$ remains odd throughout all transitions and no valid previous state leads to $(0,1)$.

6) Prove: We need to prove that strong induction is not stronger than regular induction.

Assumption: $Q(n)$: " $P(1), P(2), \dots, P(n)$ are all true."

Base case ($n=1$):

Let us assume that $P(1)$ is true. Since, we already know that $Q(1)$ states that $P(1)$ is true, this holds.

Inductive Step

Assume: Let us assume that $Q(k)$ is true for some k , such that $P(1), P(2), \dots, P(k)$ are true.

Prove: $Q(k+1)$ is true, such that $P(1), P(2), \dots, P(k), P(k+1)$ are all true.

We can say that by strong induction, that if $P(1), P(2), \dots, P(k)$ is true, then $P(k+1)$ is also true and if that is true then we can say that $P(1), P(2), \dots, P(k+1)$ are all true, that is, $Q(k+1)$ is also true.

Thus, by regular induction, $Q(n)$ is true for all n and hence we can also say that $P(n)$ is true for all n , we have proven $P(n)$ using only regular induction.

In the above steps we transformed, strong induction proof of $P(n)$ into a regular induction proof of $Q(n)$, we have shown that regular induction is just as powerful as strong induction.